Practical - 9

Secure User Registration with Database Insert, Duplicate Email Check, and Password Hashing

Problem Definition Implement secure user registration by inserting user details into MySQL using MYSQLi. Validate input on frontend and backend, check duplicate email/username, hash passwords using password_hash(), and display appropriate messages.

A. Theory

1. Is password_hash() used before storing passwords?

Yes. In process_register.php, before the INSERT query runs, the raw password is passed through password_hash($raw_password, PASSWORD_DEFAULT). The hashed version is then stored in the password_hash column — the plain text password is never saved to the database.

2. Is duplicate email/username checked?

Yes. Before inserting, a SELECT query runs on the students table:

SELECT * FROM students WHERE email = '$email' OR enrollment_id = '$enrollment'

If mysqli_num_rows() returns more than 0, the script stops and shows: "An account with this Email or Enrollment ID already exists!"

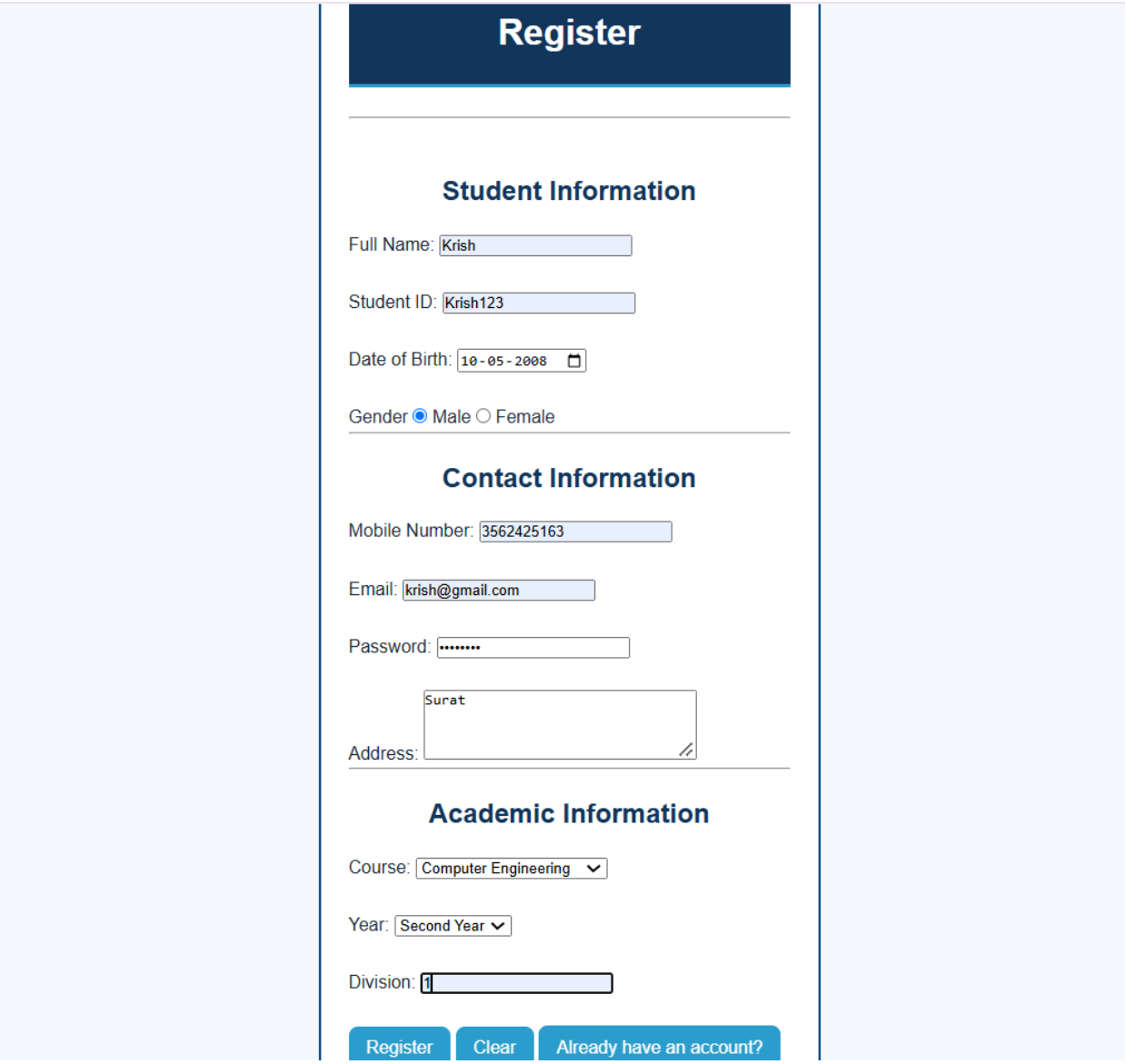
3. Are frontend and backend validations consistent?

Yes. The HTML register.html uses required attributes, type="email" for the email field, and input constraints as frontend validation. The PHP backend then performs the same checks again — empty field check, email format validation, and duplicate check — ensuring both layers are aligned.

B. Implementation

The implementation focused on secure user registration using MySQLi (basic procedural style):

C. Evidence of Work (Screenshots)



	id	student_id	email	password_hash
Edit Copy Delete	2	7	het@gmail.com	$2y$10$mpjSzAzK0nTjxZYDznH27ubyBSDh022ne9p0Ud/8Fef...
Edit Copy Delete	4	17	krish@gmail.com	$2y$10$/h0Qc4EWWrUgNGu9P7DS2.PAAbDGOZRoDOPerOmSxHo...

Check all *With selected:* Edit Copy Delete Export

stration_id	student_id	form_student_id	full_name	date_of_birth	gender	mobile	email	address	course	academic_year	division	registered_at
1	1	Niket123	Niket	2008-12-13	Male	9587248194	niket123@gmail.com	Surat	Computer Engineering	First Year	1	2026-10-19 20:37:30
3	17	Krish123	Krish	2008-05-10	Male	3562425163	krish@gmail.com	Surat	Computer Engineering	Second Year	1	2026-10-19 22:54:22

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D. GitHub Repository Link:

https://github.com/25dce037-tech/Student-Hub-Portal.git

E. Challenges Faced:

MySQLi vs PDO: Understanding the difference between procedural mysqli_query() (Practical 9) and object-oriented PDO::prepare() (Practical 8). MySQLi is simpler but less flexible.
Duplicate Check Logic: Writing a single SELECT that checks both email AND enrollment_id in one query using OR.
Password Handling: Understanding why the raw password must be passed directly to password_hash() without mysqli_real_escape_string() first — escaping can corrupt the hash.

F. Learning Outcomes:

MySQLi Procedural Connection — Learned how to use mysqli_connect(), mysqli_query(), and mysqli_num_rows() to interact with a MySQL database.
Server-Side Sanitization — Used mysqli_real_escape_string() to protect against basic SQL injection before building queries.
Secure Password Storage — Understood why plain text passwords must never be stored, and how password_hash() with PASSWORD_DEFAULT creates a secure one-way hash.
Duplicate Prevention — Implemented a pre-insert check using SELECT to prevent duplicate accounts before they cause database errors.
Form Validation (Two Layers) — Learned that frontend validation (HTML5) is for user experience, but backend validation is mandatory for security since frontend can be bypassed.

Faculty Remarks: